

Numerical and physical magnitudes are mapped into time

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In two experiments we investigated mapping of numerical and physical magnitudes with temporal order. Pairs of digits were presented sequentially for a size comparison task. An advantage for numbers presented in ascending order was found when participants were comparing the numbers' physical and numerical magnitudes. The effect was more robust for comparisons of physical size, as it was found using both *select larger* and *select smaller* instructions, while for numerical comparisons it was found only for *select larger* instructions. Varying both the digits' numerical and physical sizes resulted in a size congruity effect, indicating automatic processing of the irrelevant magnitude dimension. Temporal order and the congruency between numerical and physical magnitudes affected comparisons in an additive manner, thus suggesting that they affect different stages of the comparison process.

Keywords: Number processing; Number and time; Theory of magnitude.

The mapping of numbers into space has long been discussed. Already Galton (1880) reported that about 15% of normal adults report visuospatial representations of numbers. More recent investigators have referred to this mental spatial organization of numbers as the “mental number line” (e.g., Dehaene, 1997; Restle, 1970). Evidence for this mental number line is provided by two robust effects—the distance and the SNARC (spatial-numerical association of response code) effects.

The distance effect refers to the finding that in magnitude comparison tasks, participants' reaction times decrease as the numerical difference between the two numbers increases (Moyer & Landauer, 1967). The SNARC effect indicates that larger numbers are responded to faster with the right

hand, and smaller numbers are responded to faster with the left hand (Dehaene, Bossini, & Giraux, 1993). The most common explanation for this effect is that it reflects the mental number line, which is spatially organized with small numbers mapped onto the left side and large numbers onto the right side (see Wood, Willmes, Nuerk, & Fischer, 2008, for a review).

A different explanation for the SNARC effect was proposed by Proctor and Cho (2006). This explanation is linguistic rather than numerical in nature. Proctor and Cho expanded Clark's (1969) idea that antonymous adjectives are often asymmetrical, with one of them being the simpler unmarked linguistic form, which is more neutral and in many cases serves as the name of the

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dimension, while the other being the marked form. For example, in the long–short pair, long is the unmarked form while short is the marked one, with length being the dimension name.

Proctor and Cho (2006) argued that such asymmetry has an across-category effect through the correspondence between the markedness of different categories. This type of correspondence is demonstrated, for example, when the stimuli vary along a vertical dimension (they are located on the screen upper/lower part) and the responses along a horizontal dimension (right and left keys). In such tasks, performance is facilitated when up is mapped to a right response compared to a left response, as both right and up are the unmarked forms. According to this view, the SNARC effect is another example for the effect of polarity correspondence on classification tasks, and it is produced by the fact that both large and right are the linguistically unmarked forms.

Relatively few studies looked at the association between magnitude and time. Such an association is expected based on Walsh's ATOM (a theory of magnitude; 2003), which states that numbers, space, and time are all processed by the same magnitude system. If this is indeed the case then decisions with respect to numbers, space, and time should interact. Evidence supporting this idea was provided by studies showing that when digits were presented for variable durations, judgments of exposure duration were affected by their numerical magnitude, although such magnitude was not task relevant (Oliveri et al., 2008; Xuan, Zhang, He, & Chen, 2007).

More relevant for the present research are studies that looked at the association between numbers and time from the perspective of temporal order. Müller and Schwarz (2008) found that numerical comparisons of consecutive numbers were faster for numbers presented in an ascending (e.g., 2 followed by 3) than in a descending temporal order (e.g., 3 followed by 2). However, this effect was moderated by instructions, as it was found under *select larger* but not under *select smaller* instructions. Müller and Schwarz suggested that due to counting habits, the presentation of the first digit causes the activation of the next

consecutive digit in the form of a forward association. A similar pattern was reported by Kaan (2005) using number words, who also demonstrated this effect with event-related potentials.

Evidence that numerical magnitude is mapped into time even when numerical processing is not required by a task was provided by Schwarz and Eiselt (2009). In their temporal order judgement task, two digits were presented in rapid sequence on the two sides of a screen, and participants had to decide whether the first stimulus appeared on the left or on the right side. Sensitivity as indicated by d' was enhanced when the numerically smaller digit was presented first rather than second. This finding is consistent with automatic association between “small” and “first” and between “large” and “later”. These results however, have been recently challenged by Nicholls, Lew, Loetscher, and Yates (2011) who showed that they might be accounted for at least in part by response bias.

Together, the emerging picture is that numbers are processed faster when they appear in ascending order. This might be due to the natural expectation for magnitude to increase with time, and as a result with the mapping of small numbers with early in time and large numbers with late in time. This tendency is further enhanced by the use of label codes such as *select larger* instructions (Müller & Schwarz, 2008).

The present study

The present study extends previous research in a number of ways. First, past studies demonstrated the association or mapping between numerical magnitude and temporal order (Müller & Schwarz, 2008). In the present study, we tested the generality of the finding by examining whether it exists also for physical magnitude, and the extent to which it depends on instructions. For this purpose we constructed *pure blocks*, where participants were presented with pairs of digits that differed in their physical size for a physical comparison task and with pairs of digits that differed in their numerical size for a numerical comparison task. The digits were identical with respect to the irrelevant dimension. To tap into

the role of instructions and response codes, we used in each task both the *select larger* and the *select smaller* instructions.

Second, past studies examined the association between the processing of numerical magnitude and temporal order when there was mainly variability in the numerical dimension with minimal variability in other dimensions. In the current work, we also examined the association between magnitude and temporal order by manipulating simultaneously physical and numerical magnitudes. This was done in the *mixed blocks*, which were made of pairs of digits differing in both numerical and physical sizes. The experimental task specified the task-relevant dimension that was to be processed intentionally (e.g., numerical magnitude) and the task-irrelevant dimension that was to be ignored (e.g., physical magnitude).

It has already been well established that when two such digits are presented simultaneously for a physical magnitude comparison task, congruent trials (i.e., where the physically larger digit is numerically larger or where the physically smaller digit is numerically smaller) are processed faster than incongruent ones in which the size relations in the two magnitude dimensions are reversed (Besner & Coltheart, 1979; Henik & Tzelgov, 1982; Pansky & Algom, 2002; Tzelgov, Meyer, & Henik, 1992). This difference in performance is referred to as the size congruity effect (SiCE), and it indicates automatic processing of numerical size. The SiCE has also been reported for physical

size when participants compare numerical magnitudes (Henik & Tzelgov, 1982).

As can be seen in Figure 1, manipulating the temporal order of the task-relevant and irrelevant dimensions created the following four pair types of the mixed block: (A) pairs in ascending temporal order of both numerical and physical size; (B) pairs in ascending temporal order of numerical size, but descending temporal order of physical size; (C) pairs in ascending temporal order of physical size, but descending temporal order of numerical size; and (D) pairs in descending temporal order of both numerical and physical size. Pair Types A and D are congruent, while pair types B and C are incongruent in their numerical and physical sizes.

The mixed block provides an opportunity to examine (1) whether there is still an advantage for numbers in ascending order of the relevant dimension under conditions of variability in more than one magnitude dimension, (2) whether in a sequential presentation, the irrelevant magnitude dimension is processed automatically and thus will be reflected in a SiCE, and (3) what the relationship between the effect of temporal order of the relevant dimension and the SiCE is.

Please note that the SiCE, although reported often in numerical cognition studies, was generally found in a simultaneous presentation. The present study is the first to examine its presence in a sequential presentation. The very existence of the SiCE under such conditions of sequential

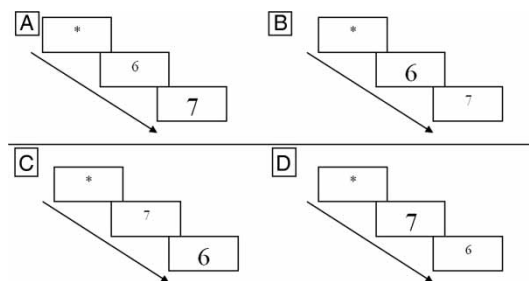


Figure 1. Examples for the four different pair types included in the mixed blocks: (A) a pair with ascending temporal order of numerical and physical magnitudes (congruent trial), (B) a pair with ascending temporal order of numerical size and descending temporal order of physical size (incongruent trial), (C) a pair with descending temporal order of numerical size and ascending temporal order of physical size (incongruent trial), and (D) a pair with descending temporal order of numerical and physical magnitudes (congruent trial).

presentation is interesting in and of itself because in a sequential presentation, the first number must be held in working memory in order to be later compared to the second. This situation raises the question of whether the participants encode both dimensions (physical and numerical sizes) of the stimulus in working memory or only the task-relevant one. The present study also provides an opportunity to test the relationship between the effect of temporal order of the relevant dimension and the SiCE on comparisons of numerical and physical magnitudes. Three possible models for such a relationship are proposed here (see Figure 2). Please note that in all the three models we assume that numerical and physical magnitudes are processed similarly. This assumption is based on the vast literature showing SiCE for comparisons of both numerical and physical magnitudes (e.g., Henik & Tzelgov, 1982; Tzelgov et al., 1992).

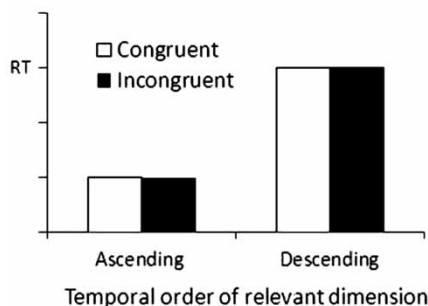
Model 1: Temporal order effect for relevant dimension and no SiCE

As mentioned earlier, the presence of a SiCE in a sequential presentation requires that both the relevant and irrelevant information of the first stimulus be kept in memory and be compared with the relevant and irrelevant information of the second stimulus. However, it is possible that for reasons of cognitive efficiency, only the relevant dimension is held in working memory. In this case there should be no SiCE, but still there should be an advantage for numbers in ascending versus descending order of the relevant dimension.

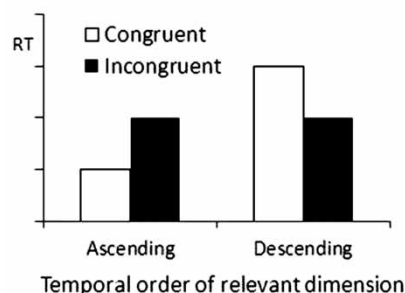
Model 2: Interactive effects of the temporal order of the relevant dimension and SiCE

This model postulates an interaction between the two factors. It is possible that the temporal order of the relevant dimension affects performance only if it is not contradicted by the temporal order of the irrelevant dimension. In such a case, the advantage for ascending temporal order will be present only if the temporal order of both dimensions converges—that is, in the case of congruent trials. Thus, Pair Type A should be processed faster than Pair Type D. In the case of

Model 1: Temporal order effect only and no SiCE



Model 2: Interactive effects of temporal order and SiCE



Model 3: Additive effects of Temporal order and SiCE

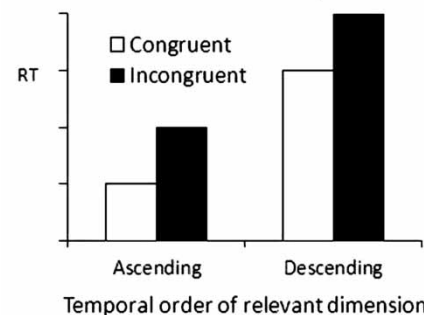


Figure 2. Predictions of the three models specifying possible relationships between effects of temporal order of the relevant dimension and SiCE (the size congruity effect). RT = reaction time.

incongruent trials, the advantage for the ascending order of one dimension is offset by the disadvantage for the descending order of the other dimension, and as a consequence there should be no effect of temporal order in incongruent trials, and the reaction time (RT) for such trials should be in between the RTs for Pair Types A and D. As can be seen in Figure 2, this model predicts an inversed congruency effect for number pairs in descending

order, and thus it implies that the effect of temporal order is stronger than that of size congruency.

Model 3: Additive effects for temporal order effect of relevant dimension and SiCE

Model 3 postulates that both effects of temporal order and effects of size congruency will be present and might affect behaviour in an additive manner. In such a case there should be both faster responses to number pairs in ascending than in descending order and faster responses to congruent than to incongruent trials.

The present study is composed of two experiments, each of which was composed of a mixed block followed by a pure block. In Experiment 1, participants performed the physical comparison task in the mixed block and the numerical comparison task in the pure block. Thus, this experiment provides information on processing of numerical magnitude, conducted both intentionally (in the pure block) and automatically (in the mixed block). Please note that the processing of the relevant dimension either in a pure block or in a mixed block is intentional because it is part of the task requirements. In the mixed block, the irrelevant dimension is thought to be processed automatically because it is not part of the task requirements (Tzelgov, 1997). In Experiment 2, participants performed the numerical comparison task in the mixed block and the physical comparison task in the pure block, and thus it completes the picture by providing information on processing of physical magnitude conducted both intentionally (in the pure block) and automatically (in the mixed block). To explore the effect of instructions, in both experiments half of the participants selected the larger number, while the other half selected the smaller number.

EXPERIMENT 1

Method

Participants

Thirty-two Psychology students participated in Experiment 1 for course credit. All were native

Hebrew speakers, with normal or corrected-to-normal vision and without learning disabilities.

Stimuli

In both tasks, the set of stimuli was composed of all possible pairs of consecutive single digit numbers (1–2, 2–3, 3–4, 4–5, 5–6, 6–7, 7–8, 8–9). Following the study of Cohen Kadosh, Henik, and Rubinsten (2008), eight physical sizes were used: 1.1°, 1.2°, 1.3°, 1.4°, 1.5°, 1.6°, 1.8°, and 2.1°. These sizes were used to create eight different pairs with two different size distances. The small distance was composed of pairs with the following physical sizes: 1.1°–1.3°, 1.2°–1.4°, 1.5°–1.8°, and 1.6°–2.1°. The large distance was composed of pairs with the following physical sizes: 1.1°–1.5°, 1.2°–1.6°, 1.3°–1.8°, and 1.4°–2.1°.

Procedure

The participants were assigned to an instruction group such that half received the instruction to select the larger number and half the smaller number. In each group, half of the participants were instructed to press the “a” key if the first number was larger (smaller) than the second number and the “l” key if the second number was larger (smaller) than the first number. The other half received an opposite mapping of responses—that is, to press the “l” key if the first number was larger (smaller) than the second number and the “a” key if the second number was larger (smaller) than the first number. Note that the instructions were the variable of interest, and the manipulation of response mapping was done for counterbalancing purposes only, and thus in the data analysis the results were collapsed across the response mappings. The mixed block of the physical comparison task preceded the pure block of the numerical comparison task to prevent activation of the numerical magnitude that might be caused by the numerical comparison task.

In the numerical comparison task, each pair was composed of two digits with the same physical size. Each pair appeared 32 times: 4 times in every physical size, and twice in ascending and twice in descending temporal order of numerical size. In the physical comparison task, each pair was composed

of pairs of different digits in different sizes. Half of them were size congruent, and the other half were size incongruent. Each pair appeared 64 times: 4 times in every physical size combination, and twice in ascending and twice in descending temporal order of numerical size. For every physical size, in every numerical order, each pair appeared twice in ascending temporal order of physical size and twice in descending temporal order of physical size. Thus, in the physical comparison task there were 512 trials, with short breaks every 128 trials, while in the numerical comparison task, there were 256 trials, with short breaks every 128 trials.

Each task began with 10 practice trials, followed by the experimental trials. In each trial a fixation cross appeared at the centre of the screen for 200 ms, followed by the first digit that appeared for 800 ms, followed by a blank screen for 300 ms. Then, the second digit appeared and stayed in view until the participant's response. The second digit appeared in the same location as the first one. The next trial began 200 ms after participant's response.

Apparatus

This and the following experiments were programmed with E-Prime software (Schneider, Eschman, & Zuccolotto, 2002) and run on IBM clone, Pentium III computers with 17-inch monitors.

Results and discussion

Two participants were excluded from the analysis due to a high error rate (>15%). Trials with incorrect answers (6%) and trials with RT shorter than 200 ms or longer than 2,500 ms (3%) were excluded from the analysis of RT. In this and in the subsequent experiment, in each task, RT and error rate (ER) were analysed separately. Significance level was defined as $p < .05$. We report partial eta-squared values (η_p^2) calculated as $SS_{\text{effect}} / (SS_{\text{effect}} + SS_{\text{error}})$.

Numerical comparison task in a pure block

Trials with the first digit being 1 or 9 were excluded from the analysis because the response could be

made prior to the presentation of the second digit. Means of RT of correct responses were submitted to a three-way analysis of variance (ANOVA) with temporal order of numerical size (ascending vs. descending) and the eight physical sizes as within-participants variables and instructions (*select smaller* vs. *select larger*) as a between-participants variable. Responses were faster for pairs in ascending (695 ms) than in descending (729 ms) temporal order of numerical size, $F(1, 29) = 5.17$, $MSE = 26,954$, $\eta_p^2 = .15$. Importantly, the interaction between temporal order of numerical size and instruction, $F(1, 29) = 3.95$, $MSE = 26,954$, $\eta_p^2 = .12$, revealed that the advantage of ascending compared to descending temporal order was present with the *select larger* instructions but not with the *select smaller* instructions (see Figure 3). No additional effects were significant, and there were no significant effects in the analysis of ER. Thus, the present results replicated those of Müller and Schwarz (2008) showing an advantage

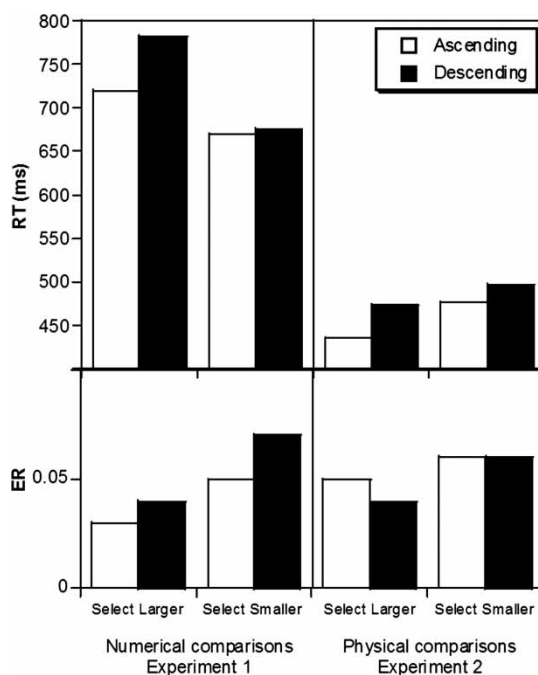


Figure 3. Mean reaction times (RTs; upper panel) and error rates (ERs; lower panel) in the pure blocks by comparison type, instructions, and temporal order.

for numbers presented in an ascending compared to descending order, which was mainly present with *select larger* instructions.

Physical comparison task in a mixed block

Means of RT of correct responses were submitted to a four-way ANOVA with congruency (congruent vs. incongruent), temporal order of the physical size (ascending vs. descending), and physical distance (small vs. large) as within-participants variables and instructions (*select smaller* vs. *select larger*) as a between-participants variable. Pairs in ascending were judged faster (686 ms) than pairs in descending temporal order of physical size (751 ms), $F(1, 29) = 18.54$, $MSE = 14,367$, $\eta_p^2 = .39$, this being insensitive to instructions (see Figure 4). Responses were faster for large physical distance (677 ms) than for small distance (760 ms), $F(1, 29) = 121.41$, $MSE = 3,503$, $\eta_p^2 = .81$. Importantly, there was also a SiCE, $F(1, 29) = 10.62$, $MSE = 2,293$, $\eta_p^2 = .27$, reflected by faster responses for congruent (709 ms) than for incongruent (728 ms) pairs. This SiCE was not modulated by physical distance. Please note

that SiCE did not interact with the effect of temporal order.

Analysis of ER also revealed a SiCE, $F(1, 29) = 6.48$, $MSE = .0018$, $\eta_p^2 = .18$. Participants made fewer errors in congruent (.06) than in incongruent (.08) trials. Participants also made fewer errors for pairs with large distance in physical size (.05) than for those with small distance of physical size (.09), $F(1, 29) = 39.73$, $MSE = .0019$, $\eta_p^2 = .58$. Also, an interaction between the SiCE and physical distance, $F(1, 29) = 10.75$, $MSE = .0014$, $\eta_p^2 = .27$, indicated fewer errors for congruent than for incongruent pairs in the small distance, but not in the large distance. Finally, the interaction between instructions and temporal order of physical size, $F(1, 29) = 6.48$, $MSE = .0047$, $\eta_p^2 = .19$, indicated that participants made more errors in ascending than in descending physical order in *select smaller* instructions, but not in *select larger* instructions.

While the advantage for ascending order of numerical size found in the present study was already well documented (Müller & Schwarz, 2008), our results expand previous ones by showing that the superiority of ascending temporal order also holds for physical magnitude. In fact, this superiority was even more robust for physical magnitude as it was found both under *select smaller* and *select larger* instructions, while for numerical magnitude it was found only under *select larger* instructions. Together the present results suggest that the association between magnitude and temporal order is a general phenomenon, consistent with ATOM.

Importantly, we also found in the mixed block a SiCE reflecting automatic processing of the irrelevant numerical magnitude. Please note that this is the first study to show a SiCE in a sequential presentation. This finding might suggest that the participants stored in working memory both dimensions (physical and numerical sizes) of the stimulus as an object file (Kahneman & Treisman, 1984), although only one dimension was task relevant.

The mixed block results show that both the effects of temporal order and those of congruency were present, and they affected behaviour in an

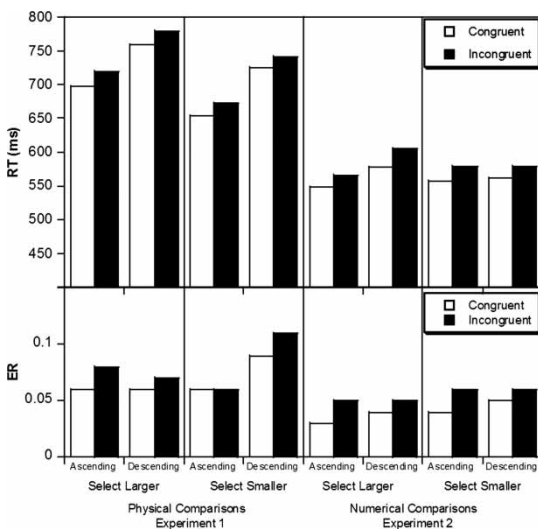


Figure 4. Mean reaction times (RTs; upper panel) and error rates (ERs; lower panel) in the mixed blocks by comparison type, instructions, temporal order of the relevant dimension, and size congruency.

additive manner. Thus, they seem to support Model 3. A further discussion of these findings is deferred until we corroborate them in Experiment 2, which included a mixed block of a numerical comparison task followed by a pure block of a physical comparison task.

EXPERIMENT 2

Method

Participants

Thirty-one new students from the same participant pool as that of Experiment 1 participated in the experiment.

Apparatus and stimuli

The apparatus and stimuli were identical to those used in Experiment 1, with the only exception that in the pure block of physical comparisons, each pair was composed of the same digit in different physical sizes. The physical sizes of Experiment 1 were used.

Procedure

The procedure was identical to the one used in Experiment 1 with the following differences. Participants first performed the numerical comparison task in a mixed block, and then the physical comparison task in a pure block. In the physical comparison task there were 288 trials, with short breaks every 144 trials, and in the numerical comparison task, there were 512 trials, with short breaks every 128 trials.

Results and discussion

Three participants were excluded from the analysis due to a high error rate (>15%). Trials with incorrect answers (4%) and trials with RT shorter than 200 ms or longer than 2,500 ms (4%) were excluded from the analysis of RT.

Physical comparison task in a pure block

Means of RT of correct responses were submitted to a four-way ANOVA with temporal order of

physical size (ascending vs. descending), physical distance (small vs. large), and nine numerical sizes as within-participants variables and instructions (*select smaller* vs. *select larger*) as a between-participants variable. Responses were faster for pairs in ascending (456 ms) than for those in descending (485 ms) temporal order of physical size, $F(1, 28) = 8.78$, $MSE = 8,552$, $\eta_p^2 = .24$. Importantly, there was no interaction between temporal order of physical size and instructions (Figure 3).

Responses were faster for pairs with large (447 ms) than for those with small physical distance (494 ms), $F(1, 28) = 153.98$, $MSE = 1,2854$, $\eta_p^2 = .85$. In addition, the effects of numerical size and temporal order of physical size interacted, $F(2, 56) = 4.10$, $MSE = 1,683$, $\eta_p^2 = .13$, such that the temporal order effect became larger as the numerical size increased.

ER analysis revealed that error rates were lower for large physical distance (.03) than for small physical distance (.07), $F(1, 28) = 16.28$, $MSE = .0074$, $\eta_p^2 = .37$. There was also a three-way interaction between instructions, numerical size, and physical distance, $F(2, 56) = 4.39$, $MSE = .0020$, $\eta_p^2 = .14$. Since this interaction was not theoretically relevant, we did not explore it further.

The main finding of the pure block in Experiment 2 is the presence of an advantage for pairs presented in ascending compared to descending temporal order of physical size. It extends the findings of Experiment 1 in the mixed block by showing that the superiority of the ascending temporal order of physical size is not constrained to the condition where the stimuli within each pair differ along both dimensions. As in Experiment 1, this advantage was not affected by mode of instructions, as it was found both under *select larger* and under *select smaller* instructions.

Numerical comparison task in a mixed block

Means of RT of correct responses were submitted to a four-way ANOVA with temporal order of numerical size (ascending vs. descending), congruity (congruent vs. incongruent), and physical distance (small vs. large) as within-participants variables and instructions (*select smaller* vs. *select larger*) as a between-participants variable. An

effect of temporal order of numerical size was found, $F(1, 28) = 8.33$, $MSE = 2,477$, $\eta_p^2 = .23$, with faster responses for ascending (562 ms) than for descending (581 ms) temporal order. Importantly, the interaction between temporal order of numerical size and instructions, $F(1, 28) = 6.48$, $MSE = 2,477$, $\eta_p^2 = .19$, indicated that the advantage of numerical comparisons of pairs presented in ascending order of numerical size was present under *select larger* instructions but not under *select smaller* instructions (Figure 4), consistent with the results in the pure block of numerical comparisons of Experiment 1.

A SiCE emerged, $F(1, 28) = 30.86$, $MSE = 877$, $\eta_p^2 = .52$, with faster responses for congruent (561 ms) than for incongruent (582 ms) trials. The SiCE was also significant in the ER analysis, $F(1, 28) = 17.13$, $MSE = .0008$, $\eta_p^2 = .38$, with lower error rates for congruent (.04) than for incongruent (.06) trials. Thus, it suggests that physical magnitude was processed automatically when participants made numerical comparisons. Again, as in Experiment 1, the temporal order effect did not interact with the SiCE.

GENERAL DISCUSSION

The following are the two novel findings of the present study. First, in comparisons both of numerical and of physical magnitude there was a consistent advantage for numbers appearing in ascending versus descending temporal order. Second, in both comparisons we observed a SiCE, indicating automatic processing of both numerical and physical magnitudes in a sequential presentation. Importantly, the effect of temporal order and that of SiCE were additive. These two findings are discussed in turn.

The advantage for comparing numbers in ascending order of magnitude

Previous studies reported an advantage for numbers in ascending temporal order when comparing numerical sizes (Kaan, 2005; Müller & Schwarz, 2008). Our results expand on previous studies by

showing that such an advantage exists not only for numerical magnitude but also for physical magnitude, consistent with the idea of a general system of magnitude (Walsh, 2003). Müller and Schwarz (2008) suggested that the advantage for numerical comparisons of pairs of consecutive numbers might be due to the “forward association”, where the presentation of a digit activates the subsequent digit, as a consequence of counting habits. Our results, showing such an advantage for physical magnitude, challenge the “forward association” as the *only* explanation of this phenomenon.

What could account for this general mapping between temporal order and magnitude? Such a mapping between time and magnitude might reflect a learning mechanism that codes the consistent correlation seen in nature between time and magnitude. For humans, animals, and plants, the increase in age is associated with an increase in physical size. This might be the basis of the participants’ expectations for magnitudes to increase with time and, as a consequence, for the facilitated performance for number pairs appearing in ascending order in either numerical or physical magnitude. Such mapping thus might originate from physical magnitude and only later be generalized to numerical magnitude. Please note that the counting algorithm can contribute to the association between numerical magnitude and time, as the counting sequence proceeds from small to large numbers, and the difference in magnitude between numbers perfectly correlates with the temporal gap between the number tags in the counting sequence. The counting algorithm might create a forward association, which might be one specific instance of a more general “later–larger” account.

The mapping of magnitude to time parallels that between magnitude and space, consistent with Walsh (2003) and Buetti and Walsh’s (2009) ATOM as well as with Lourenco and Longo (2010) who provided evidence for such a general system of magnitude for 9-month-old infants. Please note, however, that while the mapping between magnitude and space was found to depend on cultural factors, such as the direction of the writing system (Shaki, Fischer, & Petrusic, 2009; Wood et al., 2008) and on formal education

(such as the use of a number line in classes), that between magnitude and time might be more basic, as it reflects the pervasive positive relationship that exists in nature between temporal order and magnitude.

The advantage for processing numbers in ascending temporal order can also be explained through a more general mechanism, such as that of dimensional correspondence (Proctor & Cho, 2006). It might be a product of the fact that “larger” and “later” are both the linguistic unmarked forms; as a consequence, performance is facilitated when the larger number appears later. This correspondence is further enhanced when *select larger* instructions are used, since *select larger* and ascending temporal order are coded as “+”. The fact that the advantage for numbers in ascending numerical order was found more robustly for *select larger* instructions than for *select smaller* instructions fits well with this idea.

However, based on this dimensional correspondence explanation, one would predict that the pattern of results would be reversed for *select smaller* instructions, showing an advantage for numbers in descending order. However, such a pattern did not emerge. Only for numerical comparisons, the use of *select smaller* instructions eliminated the advantage for numbers in ascending order but it did not produce an advantage for numbers in descending temporal order. This finding suggests that the advantage found under *select larger* instructions was not solely due to the polarity correspondence. It should be noted that these two explanations do not contradict one another, and both factors, one that is more specific to the magnitude representation (the natural expectation for magnitude to increase) and another more general one pertaining to the dimensional correspondence, might play a role.

Interestingly, consistent across experiments and across the pure and mixed blocks, the advantage for ascending order of magnitude was found for numerical comparisons using the *select larger* instructions only, while for physical comparisons it was found using both *select larger* and *select smaller* instructions. Since this effect was not predicted, any explanation for it is a bit speculative

and post hoc. It might be a product of the fact that the association between physical magnitude and time is more basic as it originates from the life-long learning process showing that physical magnitude increases with time. The mapping between numerical magnitude and time might be secondary to the mapping between physical magnitude and time and, as a consequence, be less robust.

The presence of a SiCE

In both experiments when participants performed the numerical and physical comparison tasks, the SiCE emerged, indicating an automatic processing of the irrelevant magnitude dimension. The finding of a SiCE in a sequential presentation is novel, and it suggests that not only task-relevant information was being held in working memory but also task-irrelevant information. This finding is especially interesting when taking into consideration the fact that working memory is so limited in capacity. The storage of both relevant and irrelevant dimensions together in working memory is consistent with Kahneman and Treisman’s (1984) object file idea. Kahneman and Treisman proposed that an object file is formed very early in the processing of an object and that it includes various types of information, both information that is required by the task and information that is not. This pattern of results is also consistent with studies showing that working memory capacity is limited in terms of objects rather than features (Luck & Vogel, 1997).

The fact that temporal order of the relevant dimension and the SiCE affected behaviour in an additive manner supports Model 3 proposed in the introduction section (Figure 1). One way to interpret this additive pattern would be to suggest that they affect different stages in the comparison process. The effect of temporal order of the relevant dimension might reflect an early phase of expectations for things to increase in magnitude with time, while size congruency might reflect mainly a late stage of response competition (e.g., Cohen Kadosh et al., 2007).

The additive pattern of the SiCE and temporal order effects might also help to clarify the source of the SiCE found in the present study. Incongruent

trials (Pair Types B and C) involve incongruence in physical and numerical magnitudes but also incompatibility in temporal order of physical and numerical magnitudes. As a consequence, it is hard to tell whether a SiCE reflects mainly incongruence in magnitude dimensions or incompatibility in temporal order of the magnitudes. The fact that the temporal order effect and the SICE were additive and did not interact supports the former interpretation. Further support is provided by the fact that the effect of temporal order was moderated by instructions, at least in the case of numerical comparisons, while the SiCE was not. Moreover, the SiCE found was similar to the one found in simultaneous presentations as it was present in both numerical and physical comparison tasks. Together, these pieces of evidence suggest that the SiCE in a sequential presentation reflects an effect of incongruence in magnitude.

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